

PLAYWRIGHT → CORE CONCEPTS (COMPLETE)

We will cover:

1. Architecture
 2. Installation & Project Setup
 3. Test Runner (test, expect)
 4. Browser Context & Page
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1 PLAYWRIGHT ARCHITECTURE

1 What is it

Playwright architecture defines how tests interact with browsers using a **Node.js layer + browser engines (Chromium, Firefox, WebKit)**.

2 Key Components

- Test Script (JS/TS)
 - Playwright API
 - Browser Server
 - Browser Engines
 - Browser Context
 - Page
-

3 How YOU should answer (Interview Style)

Playwright follows a client-server architecture where test scripts communicate with browser engines through Playwright APIs. It supports multiple browsers like Chromium, Firefox, and WebKit, and uses browser contexts for isolated test execution.

4 Practical Example

```
</> JavaScript   
  
test('example', async ({ page }) => {  
  await page.goto('https://example.com');  
});
```

👉 Script → Playwright → Browser → Execution

5 Common Mistakes

- ✗ Saying “Playwright only works on Chrome”
 - ✗ Not mentioning **browser context**
 - ✗ Over-explaining internal working
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6 Expected Questions

- What is Playwright architecture?
 - How is it different from Selenium?
 - What is browser context?
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2 INSTALLATION & PROJECT SETUP

1 What is it

Setting up Playwright project using Node.js and installing required dependencies.

2 Key Components

- Node.js
 - npm
 - @playwright/test
 - Browsers installation
 - Project structure
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3 How YOU should answer

I set up Playwright using Node.js by installing @playwright/test via npm. Then I install browser binaries and create a test folder structure for writing test cases.

4 Practical Example

```
</> Bash
npm init -y
npm install -D @playwright/test
npx playwright install
```



5 Common Mistakes

- ✗ Saying "Playwright needs Selenium"
- ✗ Not knowing commands
- ✗ Not knowing project structure

6 Expected Questions

- How do you install Playwright?
 - What are prerequisites?
 - What is project structure?
-

3 TEST RUNNER (test, expect)

1 What is it

Playwright Test Runner is used to execute tests and perform assertions.

2 Key Components

- test()
 - expect()
 - hooks (beforeEach, afterEach)
 - fixtures
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3 How YOU should answer

Playwright provides a built-in test runner where test() defines test cases and expect() is used for assertions. It supports fixtures and hooks for better test management.

4 Practical Example

```
JavaScript

test('title check', async ({ page }) => {
  await page.goto('https://example.com');
  await expect(page).toHaveTitle(/Example/);
});
```

5 Common Mistakes

- ✗ Confusing test() with function
- ✗ Not knowing expect() usage
- ✗ Ignoring hooks

6 Expected Questions

- What is test() in Playwright?
- What is expect()?
- What are hooks?

4 BROWSER CONTEXT & PAGE

1 What is it

Browser context is an isolated session, and page represents a tab.

2 Key Components

- Browser
- Context
- Page

3 How YOU should answer

Browser context is like a separate session similar to incognito mode, and page represents a single tab. This helps in running tests independently without affecting each other.

4 Practical Example

```
</> JavaScript  
  
const context = await browser.newContext();  
const page = await context.newPage();
```

5 Common Mistakes

- ✗ Saying context = browser
 - ✗ Not understanding isolation
 - ✗ Confusing page and context
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6 Expected Questions

- What is browser context?
- Difference between browser and context?
- Why is context important?